

Vinícius de Paulo Souza Ribeiro

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ACADEMIC EDUCATION

Doctor of Philosophy (Ph.D.) – Université de Lorraine Oct 2020 – Dec 2023

Computer Science – Deep Learning, Speech Production

Thesis title: *Deep Supervision of the Vocal Tract Shape for Articulatory Synthesis of Speech*

Extracurricular activities:

- 12th Lisbon Machine Learning School – Instituto Superior Técnico, Lisboa, Portugal (July 2022)
- 4th International School on Deep Learning – Università Politecnica delle Marche, Universitat Rovira i Virgili & IRDTA, Las Palmas de Gran Canaria, Spain (July 2021)

Master of Science (M. Sc.) – University of Campinas Aug 2017 – Aug 2019

Electrical Engineering – Deep Learning, Computer Vision

Dissertation title: *The Impact of Annotation Quality on Deep Learning for Skin Lesion Segmentation*

Extracurricular activities:

- Summer School on Data Science for Document Analysis and Understanding – La Rochelle Université & Inria, France (July 2019)

Bachelor of Science (B. Sc.) – University of Campinas Feb 2011 – Aug 2016

Electrical Engineering

Extracurricular activities:

- Software Engineering Internship – Motorola Mobility, Jaguariúna, Brazil (2015 – 2016)
- Linear Analysis Lab Teaching Assistant – University of Campinas, Campinas, Brazil (1st Semester, 2015)
- Academic Internship – Czech Technical University, Prague, Czech Republic (Jan – Feb 2015)
- Academic Internship – University of Campinas, Campinas, Brazil – Funded by PIBITI/CNPq (2014 – 2015)

PUBLICATIONS

My work has appeared in premier international venues including *CVPR*, *Interspeech*, and *ICA*, as well as high-impact journals such as *Speech Communication* and *Computer Methods and Programs in Biomedicine*, with additional pre-print work on arXiv.

Below is a short list of relevant publications in which I am the first author. For my complete list of publications, visit my [Google Scholar](#) profile or my [personal website](#).

Automatic Segmentation of Vocal Tract Articulators in Real-Time Magnetic Resonance Imaging 2023

Vinicius Ribeiro, Karyna Isaieva, Justine Leclerc, Jacques Felblinger, Pierre-André Vuissoz, Yves Laprie

Computer Methods and Programs in Biomedicine

Autoencoder-Based Tongue Shape Estimation During Continuous Speech 2022

Vinicius Ribeiro, Yves Laprie

Interspeech 2022

Automatic Generation of the Complete Vocal Tract Shape from the Sequence of Phonemes to be Articulated 2022

Vinicius Ribeiro, Pierre-André Vuissoz, Karyna Isaieva, Justine Leclere, Yves Laprie
Speech Communication

Towards the Prediction of the Vocal Tract Shape from the Sequence of Phonemes to be Articulated 2021

Vinicius Ribeiro, Karyna Isaieva, Justine Leclere, Pierre-André Vuissoz, Yves Laprie
Interspeech 2021

LANGUAGES

Portuguese	Native
English	Fluent
French	Advanced

PROFESSIONAL EXPERIENCES

The following is a selection of my most recent positions. For a complete work history, visit my [LinkedIn](#) profile.

Intercom	Jan 2024 – Current
<i>Senior Machine Learning Scientist</i>	<i>Dublin, Ireland</i>

- Fine-tune open-source large language models via supervised fine-tuning and reinforcement learning, with a focus on building production-ready agentic AI capabilities
- Architect AI agent systems; drive model quality through iterative prompt engineering and rigorous A/B experimentation

Loria	Oct 2020 – Dec 2023
<i>Doctoral Researcher (Ph.D. Candidate)</i>	<i>Nancy, France</i>

- Led deep learning research on articulatory speech synthesis, developing neural network models trained on multimodal physiological and acoustic data
- Built models for real-time MRI segmentation, articulatory motion tracking, and speech processing
- Published peer-reviewed, state-of-the-art research at top international conferences and in high-impact journals

Meta AI	Aug 2022 – Dec 2022
<i>Research Scientist Intern</i>	<i>New York, USA</i>

- Collaborated with Meta AI's Speech Team to investigate alignment techniques for robust wake word detection
- Delivered research resulting in a first-author publication at Interspeech 2023

Dasa & Nexa Digital	Oct 2018 – Oct 2020
<i>Data Science Specialist</i>	<i>São Paulo, Brazil</i>

- Developed deep learning models for medical image analysis across CT, MRI, ultrasound, and X-ray modalities
- Built ML solutions that streamlined clinical workflows and reduced manual effort for medical teams
- Designed ETL pipelines to extract structured information from unstructured PDF medical reports
- Established data governance and traceability standards across ingestion, processing, and storage workflows